Stockout Why

An AI agent that explains why a product ran out, and cites the rows behind every claim.

Zetaris | Meterless H-MEM | Meterless Nightshift | NVIDIA Nemotron

```
PRIMARY CAUSE
PO-506 took 19 days
against a quoted 7

[supported] late vs promise
[supported] demand rose 1.20x
  ...but not sufficient
[ruled out] reorder point
[ruled out] order raised late

nemotron   yes
```



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THE PROBLEM

An empty shelf has two suspects

SUSPECT 1

The supplier was late

?

Blame demand instead and you over-order: cash tied up in stock you didn't need, while the late supplier goes unchallenged.

SUSPECT 2

Demand spiked

?

Blame the supplier instead and you start a dispute you can't prove, and the shelf empties again the next time demand jumps.

Most tools rank plausible causes. A plausible cause isn't a proven one. **Stockout Why proves which cause mattered, and shows the rows.**

WHAT IT DOES

One question in, a verdict with receipts out

1

The verdict

The primary cause, computed from the company's own records. Same data, same answer, every time.

2

The reasoning

Every candidate cause tested and marked supported or ruled out, with the numbers that decided it.

3

The evidence

The SQL, row count and result hash behind each claim, so anyone can re-run it in Zetaris.

From the live report for SKU-1041 (abridged)

```
WHY DID SKU-1041 RUN OUT?
Stock reached zero on 2026-09-03 for 8 days.

PRIMARY CAUSE
  PO-506 took 19 days against a quoted 7.
  Repairing it alone prevents the stockout.

[supported] Delivery 12 days past its promise
[supported] Demand rose 1.20x, not sufficient
[ruled out] Reorder point too low
[ruled out] Order raised late

E1  On-hand stock, SKU-1041: 180 rows, b4dc2ddd8aa8
```

HOW IT DECIDES

Counterfactuals, not weights

The agent replays the stock history with one cause repaired at a time.

The primary cause is the one whose repair alone would have prevented the stockout.

Contributing causes are real but not sufficient; the rest are ruled out, with numbers.

19 days

PO-506's lead time, against a quoted 7

Cause tested	What the records show	Verdict
Supplier took longer than its quoted lead time	19 days against a quoted 7. Arriving in 7 days (2026-08-30) avoids the stockout	Primary
Delivery missed its promised date	Promised 2026-08-30, arrived 2026-09-11: 12 days late	Contributing
Demand rose before the stockout	32.8 vs 27.5 units/day (1.20x, under the 1.5 spike line). The shelf empties even at the lower rate	Contributing
Reorder point too low	336 units covers 7 days at 32.8/day (230 needed)	Ruled out
No order raised at the reorder point	Crossed on 2026-08-23; PO-506 raised the same day	Ruled out

Four layers, one door to the data

DATA

Zetaris

Six retail tables, federated as one datasource. Seven fixed, read-only queries; every read is an evidence record.

CORE

Counterfactual replay

History re-run with one cause repaired at a time. Deterministic: no model in the verdict.

BUILD

Meterless H-MEM

Remembers past verdicts per supplier and learns from analyst feedback. 13/13 conformance.

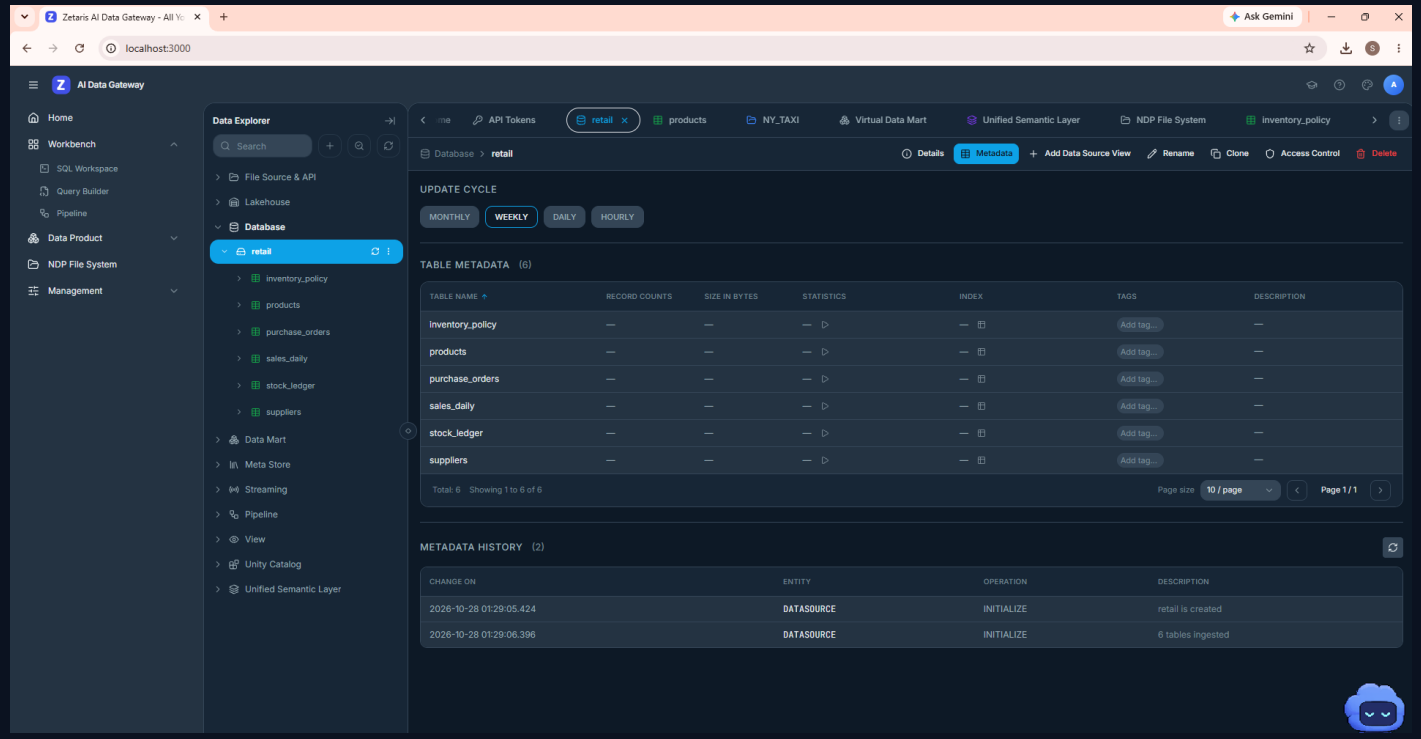
POWER

Nemotron via Nightshift

Writes the plain-words explanation, fact-checked before it is shown. Never decides.

Output verdict · reasoning · re-runnable evidence · a memory for next time

Zetaris is the only door to the data



Zetaris Data Explorer: the retail datasource and its six tables (local install)

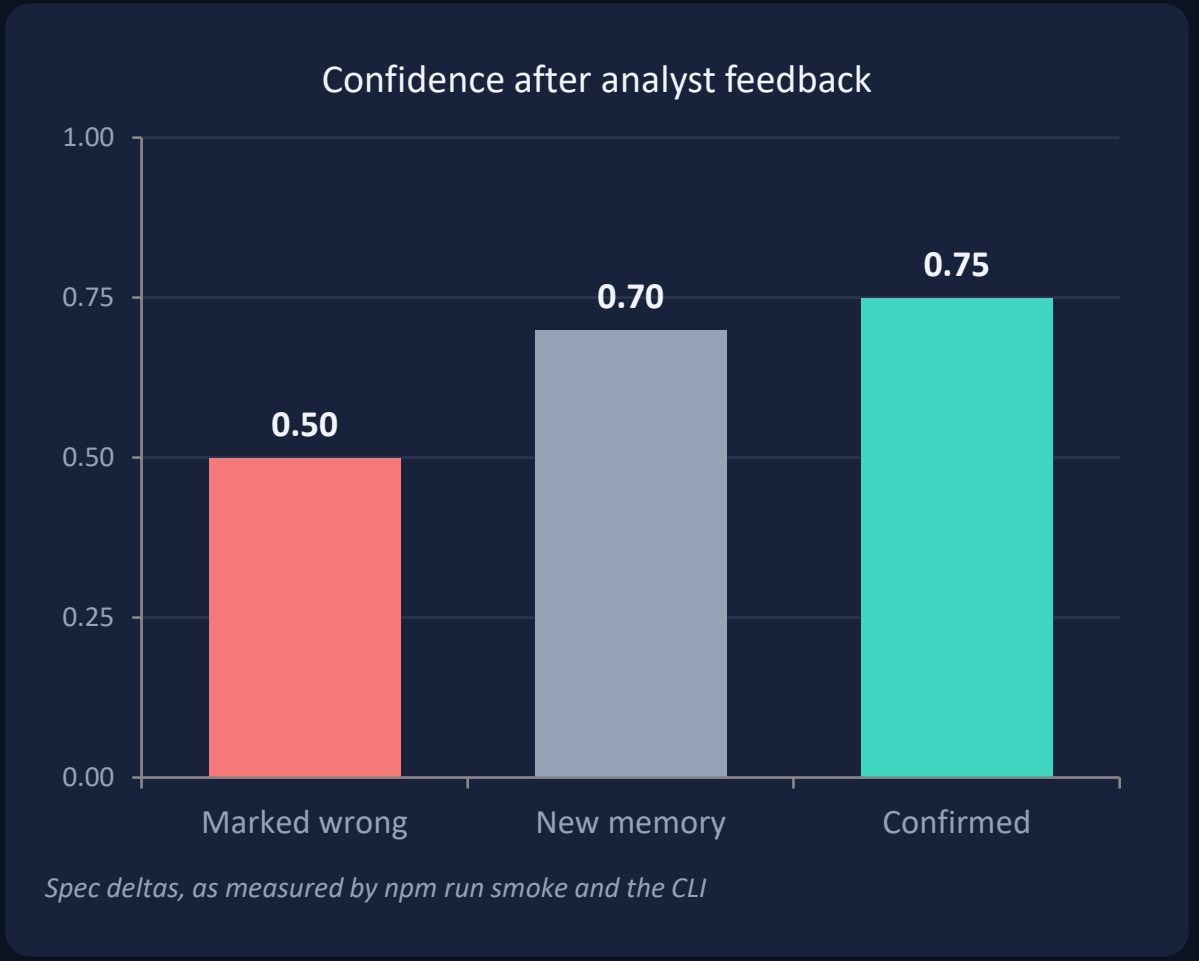
- Six tables in a retail Postgres, registered in Zetaris as the datasource retail
- Seven fixed, read-only queries. The agent chooses one; it can't write SQL
- Every read returns its SQL, rows, result hash and timestamp
- The loader counts each table back through Zetaris: all six match

Memory that compounds

13/13

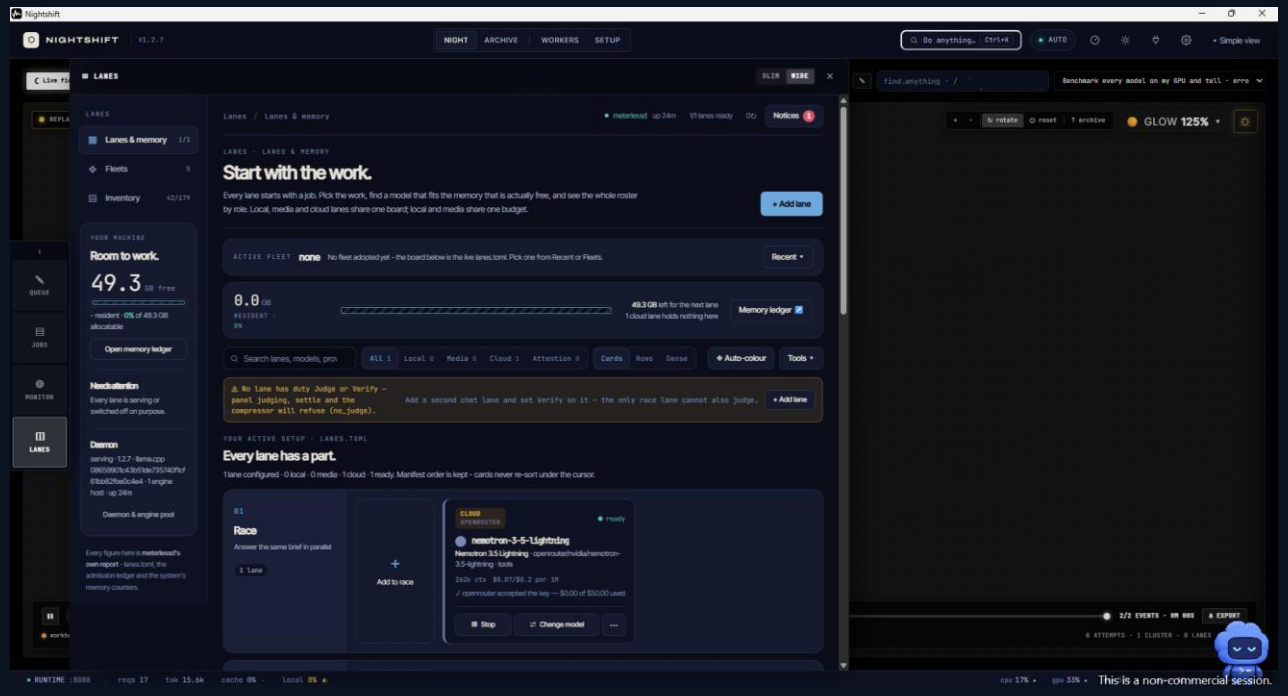
official H-MEM conformance cases, on the canonical upstream runner

- Our own implementation, written against the published spec
- Each verdict is stored with its lineage, per supplier
- The next investigation recalls it, ranked by confidence
- Analyst feedback moves confidence by the spec's exact deltas



Spec deltas, as measured by npm run smoke and the CLI

Nemotron explains; the records decide



Nightshift lane: NVIDIA Nemotron 3.5 Lightning via OpenRouter, ready

```
MODEL CALL
requested   openrouter/nvidia/nemotron-3.5-lightning
answered    openrouter/nvidia/nemotron-3.5-lightning
receipt     rcpt_req_udaamg4e
nemotron    yes
```

- Nightshift is the app's primary AI provider, over its OpenAI-compatible Runtime
- The model is pinned to Nemotron, and the report logs the model that actually answered
- Provider keys stay in Nightshift; the app holds only the Runtime URL and token

TRUST

Every model sentence is checked before it's shown

1

Claim checker

Every number and date must appear in the findings or the evidence. Dates in words, ordinals and thousands are read too.

2

Discard, don't caveat

Prose with one unsupported figure is thrown away, and the computed wording is shown instead.

3

No cut-off answers

Empty or truncated replies are retried, up to three times, and never published.

4

A bounded role

The model writes the explanation only. An outage degrades the prose, never the verdict.

“Every one of the 16 figures above was checked against the evidence before publishing.”

From the live report's explanation section

PROOF

Verified on a live stack

6/6

setup checks

npm run smoke: Zetaris, Nightshift + Nemotron, H-MEM

39/39

tests passing

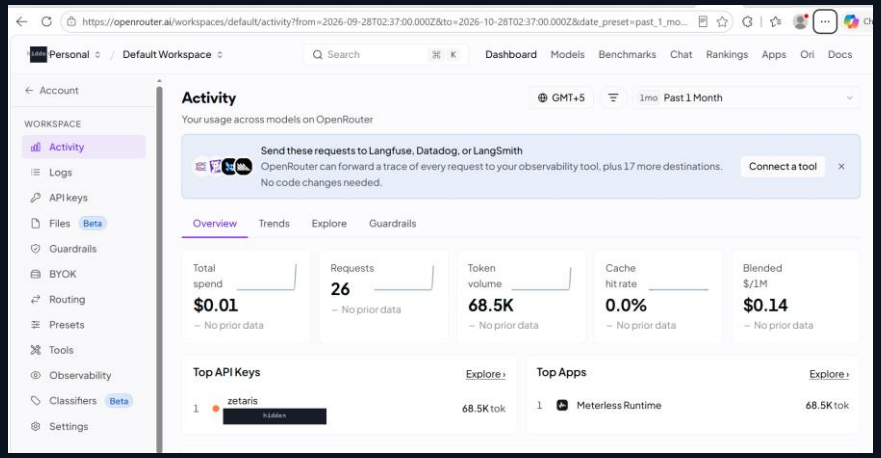
npm test, from a clean clone

13/13

H-MEM conformance

the canonical upstream runner

- Live and sample mode reach the same verdict, word for word
- All six tables count back exactly through Zetaris
- Nemotron answers through Nightshift, every figure verified



HONEST LIMITS

What we don't claim

! Synthetic data

180 days simulated day by day, with one injected fault and one distractor. No claim about a real retailer's books.

! One true cause exercised

All five causes are tested, but only the lead-time slip is exercised as the true cause.

! Local Zetaris only

The live path is verified against Zetaris running in Docker, not a hosted instance.

! Simple recall

H-MEM's embedder is a bag-of-words projection: it matches shared vocabulary, not paraphrase.

! Figures, not wording

The claim checker stops invented numbers, not a correct number in a loosely worded sentence.

! Variable latency

Nemotron via OpenRouter takes 6 s to about 2 minutes, depending on the upstream provider. The app retries.

TRY IT

Run it yourself

Sample mode: no keys, no network

```
$ npm install
$ npm run gen-data
$ npm run investigate -- --sku SKU-1041 --sample
$ npm test
```

Live mode: Zetaris + Nightshift

```
$ cp .env.example .env
$ npx tsx scripts/register_zetaris.ts --execute
$ npm run smoke
$ npm run investigate -- --sku SKU-1041 --explain
```

Code: github.com/SohaibAamir28/stockout-why

Needs Node 22. Live mode also needs a local Zetaris install and Meterless Nightshift; provider keys go in Nightshift's Providers tab, never in .env.

BUILT BY

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MS Computer Science, NUST (2025–2027)

“I build AI systems that have to work on real, messy data, and I measure where they fail.”

- 1st place, AI for Connectivity Hackathon 2025
- Global Nominee, NASA Space Apps 2024
- All 9 puzzles solved, Harvard CS50x Puzzle Day 2025

linkedin.com/in/sohaib-aamir · github.com/SohaibAamir28/stockout-why

Stockout Why: the agent that can explain why.